From Censorship to Detoxification: Rewriting Harmful Memes

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Abstract

Harmful memes are often toxic because image and text jointly imply an identity-based attack. Instead of detecting and removing such content, we study text detoxification: given a meme image and caption, produce a safer caption that preserves the topic and image-text coherence. Our system uses LLaVA-NeXT as a multimodal teacher to generate structured explanations and pseudo-rewrites, then distills this behavior into a LoRA-adapted BART-large student. The main deployment contribution is a CLIP Proxy + BART path that maps CLIP image/text features to BART encoder soft tokens, avoiding a large VLM/LLM at inference. On 280 held-out memes, the full BART student reaches 0.963 ± 0.006 Text STA and 0.207 ± 0.006 STA delta across five seeds, compared with the single reported DetoxLLM-7B baseline at 0.940 and 0.184. The proxy obtains the highest BERTScore and CLIPScore (0.581 ± 0.020, 0.638 ± 0.002), suggesting stronger source retention and image alignment, but lower Text STA shows weaker toxicity control.

Keywords: meme detoxification, multimodal learning, text rewriting, distillation, CLIP, BART.

1. Introduction

Hateful meme moderation is usually framed as classification: predict whether an image-text pair is harmful and then remove or suppress it [1, 2]. This is necessary but incomplete. Meme captions can be ambiguous alone, while the image supplies the target, stereotype, or implied insult. A pure text detoxifier can miss that interaction; a pure detector does not preserve a benign version of the communicative intent.

We formulate harmful meme moderation as *text rewriting*. Given a meme image and overlaid text, the goal is to rewrite the text so that the harmful framing is reduced while the topic, short meme style, and visual compatibility remain recognizable. This raises two questions. First, can multimodal teacher explanations supervise a compact meme rewriter? Second, can CLIP-derived soft encoder states provide a VLM-free inference path that keeps useful visual grounding without running a large generator?

Our contributions are threefold. First, we implement an explain-then-rewrite pipeline that uses LLaVA-NeXT [3] to produce target groups, visual evidence, implicit meanings, and pseudo-detoxified captions. Second, we fine-tune BART-large [4] with LoRA [5] on these meme-specific pseudo-labels and compare it to DetoxLLM-7B [6] and non-fine-tuned BART. Third, we introduce a deployment-oriented CLIP Proxy + BART path: CLIP [7] encodes the image and text, a small MLP predicts BART encoder-memory soft tokens, and the fine-tuned BART decoder generates the rewrite without running LLaVA or another 7B language model at inference.

2. Related Work

Hateful meme understanding. The Hateful Memes benchmark showed that multimodal reasoning is required because unimodal cues are deliberately confounded [1]. VisualBERT [2] and CLIP [7] provide useful image-text representations, but the standard objective is still detection. Our work differs by producing a revised caption rather than only a moderation label.

Text detoxification. ParaDetox casts detoxification as parallel style transfer for toxic sentences [8], and DetoxLLM trains large causal models for detoxification with explanations [6]. These methods are primarily text-only at inference. We use them as the closest external task baseline, but our supervision is meme-specific and grounded in teacher-generated multimodal explanations.

Multimodal teachers and efficient students. LLaVA-NeXT improves visual instruction following and OCR-oriented reasoning [3], which makes it suitable for teacher pseudo-labeling. However, using a 7B VLM at deployment is costly. We therefore distill the teacher into BART-large and test whether CLIP-derived soft encoder states can preserve useful visual context. The gap is the combination: detection predicts labels but does not rewrite, text-only detoxification ignores image-caption interactions, and direct VLM rewriting is expensive at inference. Our method uses multimodal teacher supervision while removing the large VLM/LLM from deployment.

3. Method

Data construction. We combine HarMeme [9], MAMI [10], and MMHS150K [11]. Stage 0 extracts text with EasyOCR and keeps images with 10–300 OCR characters. CLIP then filters non-meme images: HarMeme and MAMI use a binary meme-vs.-screenshot comparison, while MMHS150K uses a five-prompt rule and keeps an image only if the meme prompt is top and has probability at least 0.45. Filtered examples are split 80/10/10, stratified by dataset and harmful label.

Teacher pseudo-labels. For each split, LLaVA-NeXT first generates an explanation with three fields: target group, visual evidence, and implicit harmful meaning. A second prompt asks for a short non-hateful rewrite that keeps the meme topic and style. Stage 1 records all attempts but marks pass/fail using text STA, BERTScore, and toxicity-drop filters. The Stage 2 builder keeps rows that passed Stage 1, have no parse error, have positive toxicity reduction, pass rewrite-quality checks, and have normalized source/rewrite similarity at most 0.95. The final supervised data contains 3,578 train, 269 validation, and 280 held-out test examples.

BART student. The student starts from BART-large and is trained on the meme pseudo-rewrite data. Encoder inputs contain the original text plus optional teacher fields in an explicit detoxification instruction. We train four conditions: `full`, `target_only`, `visual_only`, and `none`. LoRA is applied to BART attention projections and feed-forward layers with rank 32, alpha 64, dropout 0.05, learning rate 10^{-4}, batch size 8, warm-up 50 steps, weight decay 0.01, and 5 epochs. About 17M of roughly 400M parameters are trainable.

CLIP Proxy + BART. The proxy is a 3-layer MLP. It receives concatenated CLIP image and text embeddings (1536 dimensions) and predicts a sequence of $K = 16$ BART encoder soft tokens of size 1024. During training, the target is the pooled encoder memory of the full-condition fine-tuned BART. At inference, proxy tokens are concatenated with BART encoder states from a text-only null-context prompt, then decoded by the fine-tuned BART decoder. The architecture is illustrated in Figure 1. This turns multimodal grounding into a continuous prefix for the decoder rather than requiring a textual explanation from a VLM at inference. It is our main deployment path because it keeps visual input through CLIP while removing the LLaVA teacher and DetoxLLM-sized causal LM from inference.

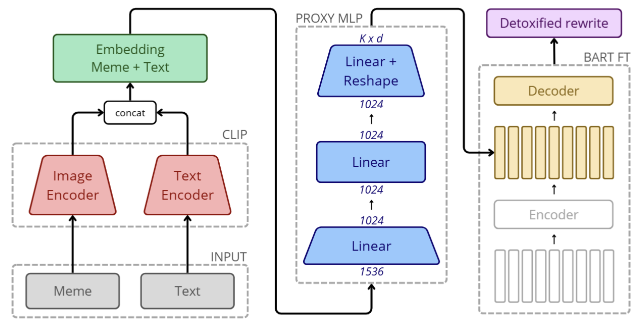

Figure 1. Efficient inference path: the proxy maps CLIP meme/text embeddings to BART soft encoder tokens for detoxified decoding.

4. Data Licensing and Ethics

The project uses only public or research-access datasets already documented in the repository materials. Those materials identify HarMeme and MAMI as CC 4.0 research datasets; the pipeline downloads HarMeme automatically, documents MAMI as a manual-request dataset, and uses MMHS150K through the authors' research download links rather than redistributing it. The paper reports aggregate metrics and does not reproduce harmful images or captions.

Ethically, detoxification should be read as a harm-reduction tool, not as an automatic moderation authority. Rewriting can remove slurs while still preserving stereotypes, or it can over-neutralize legitimate counterspeech and satire. Toxicity classifiers and hate datasets also encode annotation and platform biases. For deployment, human review, uncertainty reporting, and clear separation between educational research and real moderation decisions would be required.

5. Experiments and Results

All systems are evaluated on the same 280 held-out test examples with the same scoring pipeline and exact generated outputs: no manual correction or post-hoc sanitization is applied before scoring. Baselines are the fixed LLaVA teacher pseudo-rewrites, a single reported DetoxLLM-7B run, and base BART. Metrics are Text STA (mean non-toxic probability from a RoBERTa classifier), STA delta (rewrite STA minus original STA), BERTScore F1, and normalized CLIPScore. Higher is better for all metrics. BART-FT results are averaged across five seeds with different random initializations of the LoRA parameters, while LLaVA and DetoxLLM are single runs because of computational cost. The proxy is trained on the same five BART-FT seeds and evaluated on the same test set, so its mean and standard deviation reflect the underlying BART-FT variance rather than separate random seeds for the proxy itself.

Table 1 shows that, across five BART seeds, all BART-FT variants obtain higher mean Text STA and STA delta than

System	Text STA	STA Δ	SIM	CLIP
LLaVA teacher*	**0.990**	**0.234**	0.472	0.636
DetoxLLM-7B†	0.940	0.184	0.443	0.634
BART-FT full	0.963±0.006	0.207±0.006	0.330±0.013	0.628±0.001
BART-FT target	0.960±0.005	0.204±0.005	0.335±0.013	0.629±0.002
BART-FT visual	0.960±0.004	0.204±0.004	0.284±0.080	0.624±0.007
BART-FT none	0.957±0.013	0.201±0.013	0.325±0.030	0.628±0.002
Proxy+BART	0.884±0.015	0.128±0.015	**0.581±0.020**	**0.638±0.002**

Table 1. Test results on 280 memes. Rows with ± are seed 1–5 mean±sample std; *LLaVA is a fixed teacher target and †DetoxLLM is a single reported run.

the single reported DetoxLLM baseline. The `full` condition has the best average BART-FT detoxification, while `target_only` is close and has the best BART-only SIM. The ablation study suggests that the conditioning is not crucial for detoxification.

The external comparison and base-model ablation support the need for meme-specific fine-tuning. Base BART reaches high non-toxicity in the full condition (0.973 text STA) but has negative SIM (-0.061), indicating generic safe text rather than faithful rewriting. DetoxLLM keeps higher SIM than the BART-only students (0.443 vs. roughly 0.33), but copying can inflate BERTScore when toxic input is retained. Similarity must therefore be interpreted together with Text STA and STA delta, not as a stand-alone quality score.

The proxy exposes the deployment trade-off. It obtains the best mean SIM (0.581) and CLIPScore (0.638), suggesting stronger source retention and image alignment, but its Text STA drops to 0.884. This can partly reflect retention of harmful content, so the proxy is not the strongest detoxifier. Its advantage is architectural: it uses CLIP plus a small MLP and BART decoder, not LLaVA. In our A100 benchmark, BART-FT requires 122 ms per rewrite, compared with 9.38 s for the LLaVA teacher and 1.99 s for DetoxLLM. The proxy evaluation took 3.4 minutes for 280 examples, about 0.73 s/image, roughly 13× faster than LLaVA and 2.7× faster than DetoxLLM.

Limitations. The project depends on LLaVA pseudo-labels rather than human detoxification references, and the heterogeneous source datasets remain noisy even after filtering. We do not include human evaluation; automatic toxicity, BERTScore, and CLIPScore may miss pragmatic harm, counterspeech, satire, or cases where lexical similarity preserves the harmful implication. The BART student without the proxy receives visual context only as teacher text fields, not pixels. Finally, the proxy is not yet a drop-in replacement: it preserves similarity and alignment, but its lower Text STA requires stronger toxicity control.

6. Conclusion

We reframed harmful meme moderation as constructive text rewriting. A LLaVA-NeXT teacher supplies multimodal explanations and pseudo-rewrites, while a LoRA-adapted BART-large student is the strongest detoxification student in our evaluation relative to the single reported DetoxLLM-7B baseline. The CLIP Proxy + BART path is the novel efficient deployment route: it removes large VLM/LLM inference and improves fidelity metrics, but future work must improve proxy toxicity control while preserving efficiency and image alignment.

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